

Re: Application for the EngD position, Developing an Interactive Reverse Logistics Map for Circular Construction (CIRCOLOGIC), University of Twente

Dear Professor Marc & Selection Team,

I am writing to apply for the EngD position on the interactive reverse logistics map for circular construction (CIRCOLOGIC). This is a topic I really care about. In construction we are already good at taking buildings apart, but a lot of good material still ends up as waste. The main problem is information. People often do not know which materials are available, in what condition, where they are, or when they will need them. So they choose new materials instead of reused ones. A map that shows what is available, a way to calculate cost and CO2, and a simple way to compare options would make reuse a much easier choice. That is the kind of solution I would like to help build.

While preparing this application, I read some of Professor Voordijk's work on recovering building elements for reuse. One idea really stayed with me: reuse does not work because of technology alone, but because of how well all the parties work together, from the donor building where the material comes from to the target building where it is used again. That is close to the thinking behind my master's thesis, **Agile Project Management in the Construction Industry**. In that work I looked at how Agile can improve collaboration, continuous risk management, adaptive decision-making and resource efficiency. Completing it also showed me that better project delivery and less waste during construction only solve one stage of sustainability. This naturally led me to a broader question: what happens to materials when buildings reach the end of their service life?

Looking into circular construction, I learned that reverse logistics matters a lot, and that reuse often fails not only because of technology, but because information is split across the value chain. Details about availability, quality, ownership, location, transport and reuse options are often incomplete or not shared between stakeholders. Without reliable information, good materials may never return to construction. That is why I want to move from running projects to helping build the digital solutions that bring the right information to the right people.

My recent jobs also prepare me for this project. At Aug.e I worked on digital twins for smart buildings. We used a digital environment to understand and improve how a real building performs. For CIRCOLOGIC I see the same idea, but applied to materials: where they are, what can be reused, who is involved, and how they move through reverse logistics. At Hive CPQ I managed large manufacturer datasets and turned complex product information into simple digital solutions that non-technical users could work with. That is close to what CIRCOLOGIC needs: organising material, location and stakeholder information, connecting data from many parties, and making a reverse-logistics map easy for non-technical people to use. At iostiring I worked on a circular construction hypothesis, looking at how Agile project management can bring circular practices into real delivery. With my civil and environmental engineering background and my GIS skills, I can help build this solution and make sure people will really use it.

For the last while I have taken a break from work to take care of my newborns. During this time I kept learning on my own about circular construction, reverse logistics and GIS. This EngD is the perfect way for me to come back to work. I can use what I already know, learn more, and this time apply it to a real project in industry and not only in theory.

I would really like to do this at the University of Twente. I like that the university works closely with industry, and that the EngD combines study with a real design project together with partners like Dura Vermeer and TNO. I learn best when I can use new methods straight away and improve them with the people I work with.

I would be very happy to work on CIRCOLOGIC and to grow as an engineer who builds solutions that people actually use. I would be glad to talk more about how I can fit into the project. Thank you for reading my application.

Yours sincerely,

Pooja Awasare